# Computer Vision Assignment

## Manual SIFT Feature Detection and Description

**Course Code:** CS-482 / CV **Total Marks:** 22 Marks **Format:** Written Assignment

**Instructions:** Solve all questions by hand. Show all intermediate calculations, formulas, matrices, and diagrams. Do not use Python, MATLAB, or any other automated software. Clearly write every algebraic step on physical paper.

### Question 1: Image Gradient Calculation — 6 Marks

Consider the following 4 × 4 grayscale image patch extracted from a 16 × 16 image:

$$I = \begin{vmatrix} 10 & 10 & 10 & 10 \\ 10 & 20 & 30 & 40 \\ 10 & 30 & 50 & 70 \\ 10 & 40 & 70 & 100 \end{vmatrix}$$

Using the central difference method, calculate the horizontal and vertical gradients at pixel $I(2,2) = 20$.

$$G_x = [\, I(x+1, y) - I(x-1, y) \,] / 2 \qquad G_y = [\, I(x, y+1) - I(x, y-1) \,] / 2$$

a. **Calculate** $G_x$ and $G_y$ at pixel $I(2,2)$.

b. **Calculate** the gradient magnitude: $M = \sqrt{(G_x^2 + G_y^2)}$.

c. **Calculate** the gradient orientation: $\theta = \tan^{-1}(G_y / G_x)$.

d. **Interpret** the calculated gradient magnitude and orientation.

### Question 2: Difference of Gaussians (DoG) — 5 Marks

A small image region at two different Gaussian scales is represented by the following matrices:

$$L_\sigma = \begin{vmatrix} 10 & 20 & 10 \\ 20 & 80 & 20 \\ 10 & 20 & 10 \end{vmatrix} \qquad L_{k\sigma} = \begin{vmatrix} 15 & 25 & 15 \\ 25 & 60 & 25 \\ 15 & 25 & 15 \end{vmatrix}$$

Calculate the Difference of Gaussians matrix using: $D(x, y, \sigma) = L_{k\sigma}(x, y) - L_\sigma(x, y)$.

a. **Calculate** the complete 3 × 3 DoG matrix.

b. **Identify** the pixel with the largest absolute DoG value.

c. **Explain** whether this pixel is a potential candidate for further keypoint testing.

d. **State** why SIFT compares DoG values across neighboring pixels and scales before accepting a keypoint.

## Question 3: Keypoint Detection Using a 3 × 3 Neighborhood

5 Marks

Suppose the DoG values in a local neighborhood are:

$$D = \begin{vmatrix} 5 & 8 & 6 \\ 7 & 15 & 9 \\ 4 & 10 & 3 \end{vmatrix}$$

The center pixel has value 15.

a. **Identify** the center pixel and its eight spatial neighbors.

b. **Determine** whether the center pixel is a local maximum.

c. **Explain** why the center pixel cannot be confirmed as a final SIFT keypoint using this matrix alone.

d. In the full SIFT algorithm, **how many** neighboring samples are compared with a candidate pixel across the current, lower, and upper DoG scales?

## Question 4: Gradient Orientation Histogram

6 Marks

Suppose four pixels in a local image region have the following gradient values:

| Pixel | $G_x$ | $G_y$ |
|---|---|---|
| $P_1$ | 3 | 4 |
| $P_2$ | 0 | 5 |
| $P_3$ | −4 | 0 |
| $P_4$ | 3 | −3 |

For each pixel, calculate:

a. **Gradient magnitude:** $M = \sqrt{(G_x^2 + G_y^2)}$.

b. **Gradient orientation in degrees**, using the correct quadrant.

c. **Assign** each orientation to one of eight histogram bins, each spanning 45°, over the range 0° to 360°.

d. **Construct** the orientation histogram by adding the gradient magnitudes to their corresponding bins. Ignore Gaussian weighting and interpolation for this exercise.

**Submission Reminder:** Show complete working for every calculation. Clearly label matrices, coordinate indexes, gradient directions, and histogram bins on your handwritten submission sheets.